

Exercise 5: Configuring the Spring IoC Container

BookRepository.java:

```
package com.library.repository;

public class BookRepository {
    public void save() {
        System.out.println("BookRepository: Book saved successfully to the database.");
    }
}
```

BookService.java:

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {
    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void processBook() {
        System.out.println("BookService: Processing book logic...");
        bookRepository.save();
    }
}
```

applicationContext.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
                           http://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="bookRepository" class="com.library.repository.BookRepository" />

    <bean id="bookService" class="com.library.service.BookService">
```

```

        <property name="bookRepository" ref="bookRepository" />
    </bean>

</beans>

```

LibraryApplication.java:

```

package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;

public class LibraryApplication {
    public static void main(String[] args) {
        ApplicationContext context = new
ClassPathXmlApplicationContext("applicationContext.xml");
        BookService bookService = (BookService) context.getBean("bookService");
        bookService.processBook();
    }
}

```

Output:

```

adityahargunani@Adityas-MacBook librarymanagement % mvn clean compile
[INFO] --- resources:3.0.2:resources (default-resources) @ librarymanagement ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Copying 1 resource
[INFO] --- compiler:3.8.0:compile (default-compile) @ librarymanagement ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 4 source files to /Users/adityahargunani/Downloads/Exercise 5- Configuring the Spring IoC Container/librarymanagement/target/classes
[INFO] BUILD SUCCESS
[INFO] Total time: 0.580 s
[INFO] Finished at: 2026-06-25T13:34:10+05:30
[INFO]
● adityahargunani@Adityas-MacBook librarymanagement % mvn exec:java -Dexec.mainClass="com.library.LibraryApplication"
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 48 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB at 58 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/exec-maven-plugin/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/exec-maven-plugin/maven-metadata.xml (1.1 kB at 17 kB/s)
[INFO]
[INFO] -----< com.example:librarymanagement >-----
[INFO] Building librarymanagement 1.4
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.6.3:java (default-cli) @ librarymanagement ---
BookService: Processing book logic...
BookRepository: Book saved successfully to the database.
[INFO] BUILD SUCCESS
[INFO] Total time: 0.852 s
[INFO] Finished at: 2026-06-25T13:34:18+05:30
[INFO]
○ adityahargunani@Adityas-MacBook librarymanagement %

```

